Simulation of the cosmic rays pass through the human body with GEANT4

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Outline

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- Secondary cosmic rays at sea level
- Muon energy loss in high energy
- Simulation in GEANT4
- Some results of the present work
- Next step
- Summary

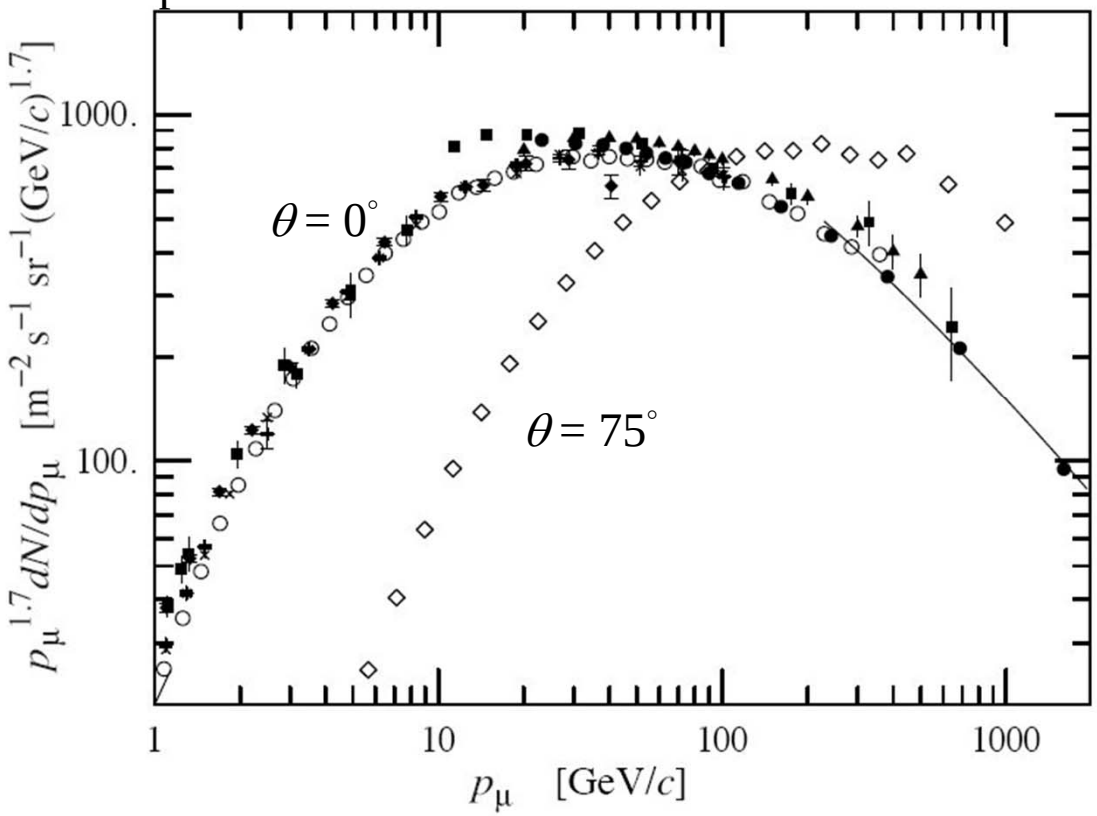
Introduction

- GEANT4 (GEometry ANd Tracking) is a software toolkit for the simulation of the passage of particles through matter by using Monte Carlo method
- It developed by CERN for high energy physics experiments
- Now it is used in high energy physics and astrophysics community and also widely used it to simulate the dosage for particle treatment in medical community
- There is visualization with GEANT4 for us to see the detector geometry and particle tracking

Introduction (continued)

- The "primary" cosmic rays are those particles accelerated at astrophysical sources, and "secondaries" are those particles produced in interaction of the primaries with atmosphere
- We want to know what the influence of the cosmic rays everywhere about the human being is

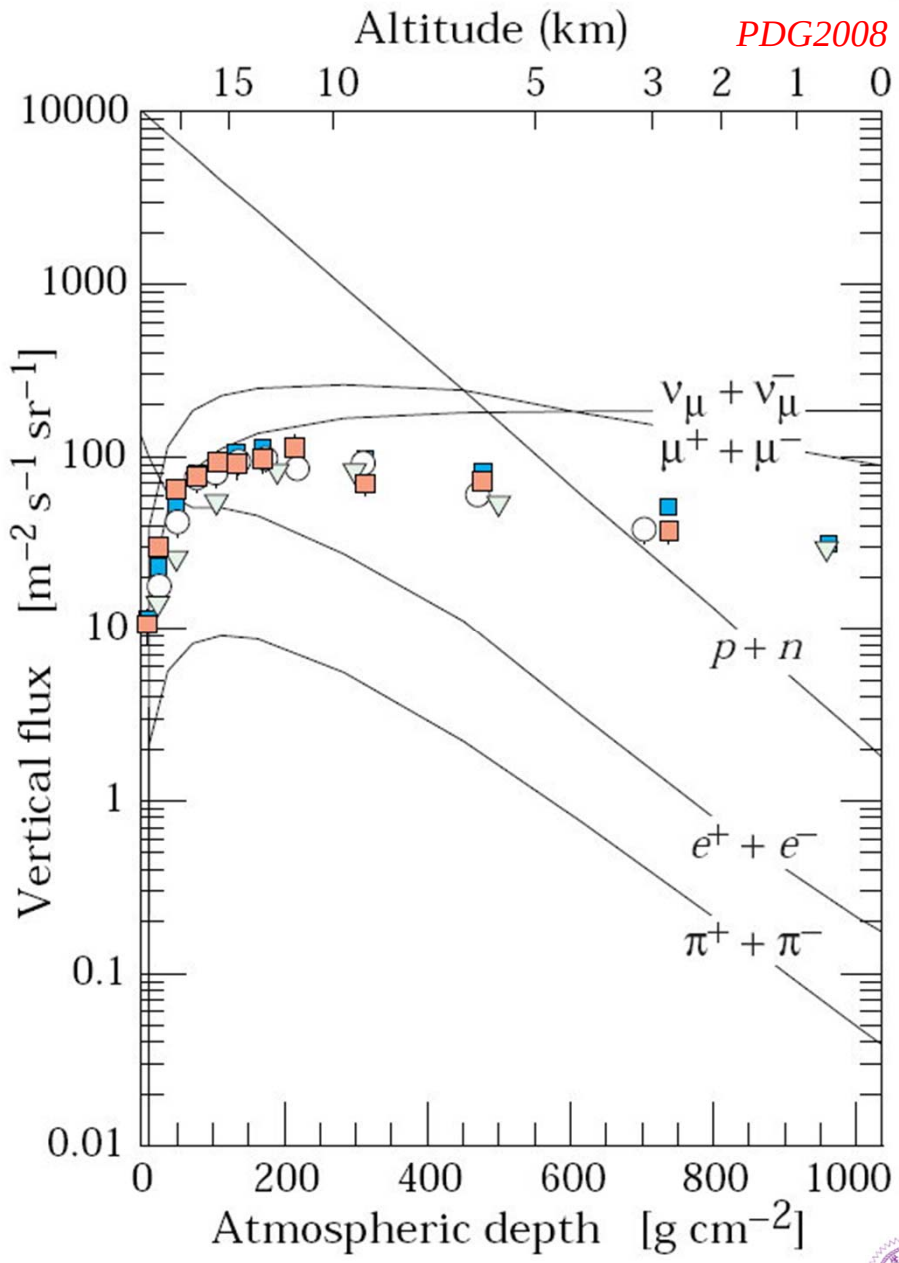
Spectrum of muons at $\theta = 0^\circ$ and $\theta = 75^\circ$ *PDG2008*



Secondary cosmic rays at sea level

- Muons are the most numerous charged particles at sea level

- The most common particle passing through human body is atmospheric muons, which are secondary particles produced high from cosmic ray interaction in atmosphere.



Muon energy loss at high energy

■ Average rate of muon energy loss as

$$-\frac{dE}{dx} = a(E) + b(E)E$$

where $a(E)$ is the mean rate of ionization energy loss given by Bethe-Bloch equation

$b(E)$ is the sum of e^+e^- pair production, bremsstralung, and photonuclear contribution

$$a(E) \equiv Kz^2 \frac{Z}{A}\frac{1}{\beta^2}\left[\frac{1}{2}\ln\frac{2m_e c^2\beta^2\gamma^2 T_{\max}}{I^2} - \beta^2 - \frac{\delta(\beta\gamma)}{2}\right], \quad \frac{K}{A} \equiv \frac{4\pi N_A r_e^2 m_e c^2}{A}$$

where

N_A is Avogadro's number

A is atomic mass of absorber

$r_e = \dfrac{e^2}{4\pi\varepsilon_0 m_e c^2}$ is classical electro radius

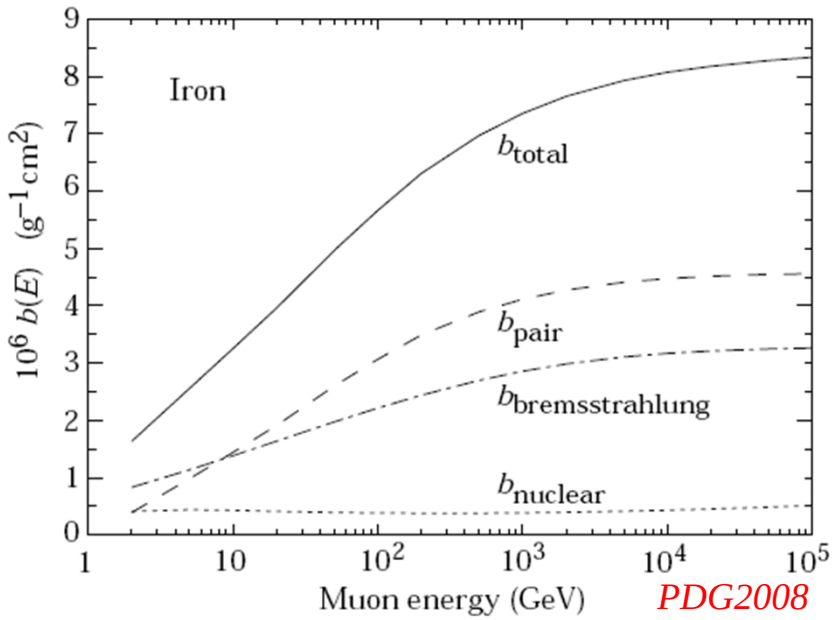
Z is atomic number of absorber

ze is the charge of incident particle

$T_{\max}$ is the maximum kinetic energy

I is mean exitation energy

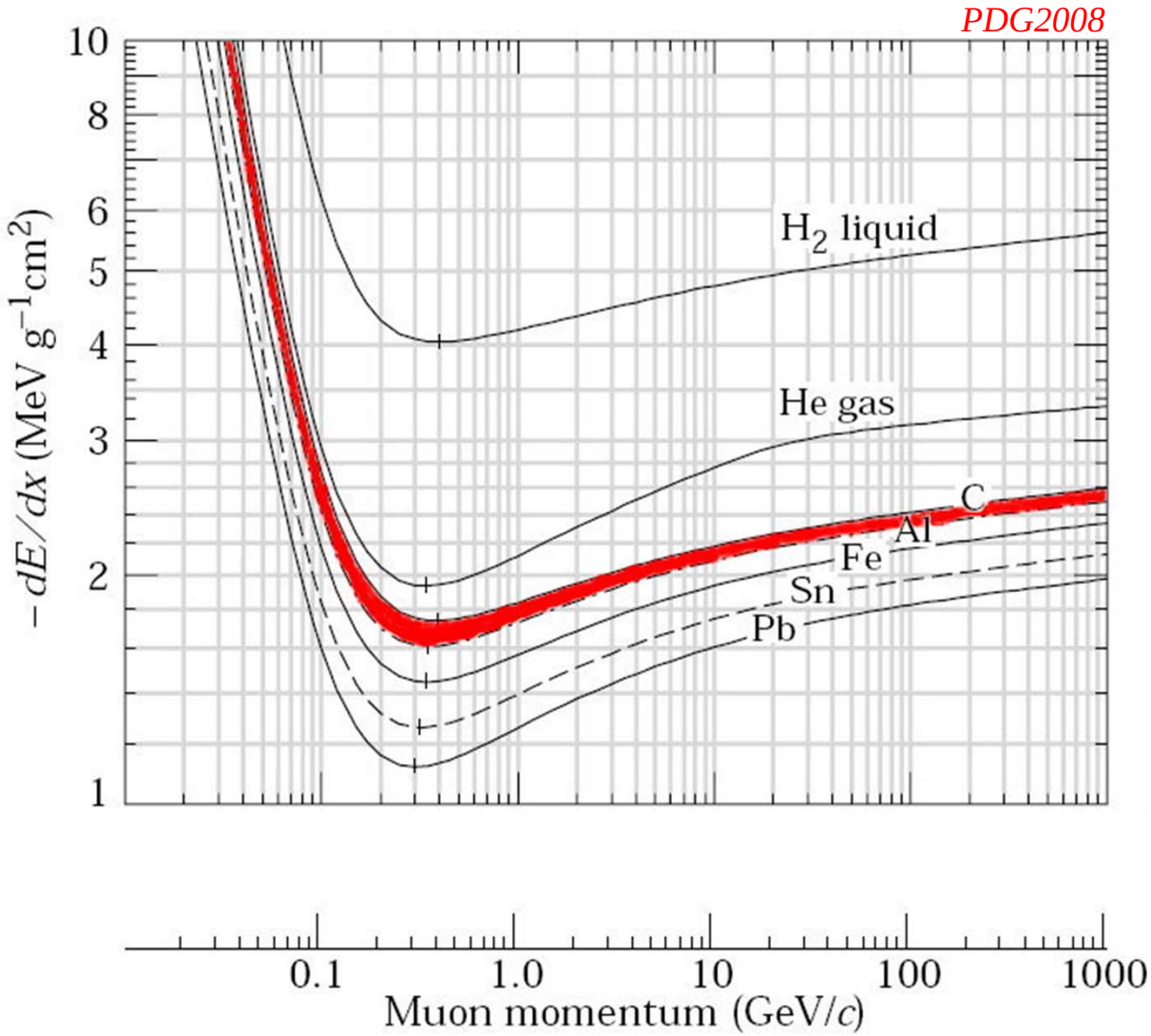
$\delta(\beta\gamma)$ is density effect corretion to ionization energy loss



PDG2008

Muon energy loss at high energy (continued)

- Energy loss in the human body
  - There is 70% H_2O in the human body



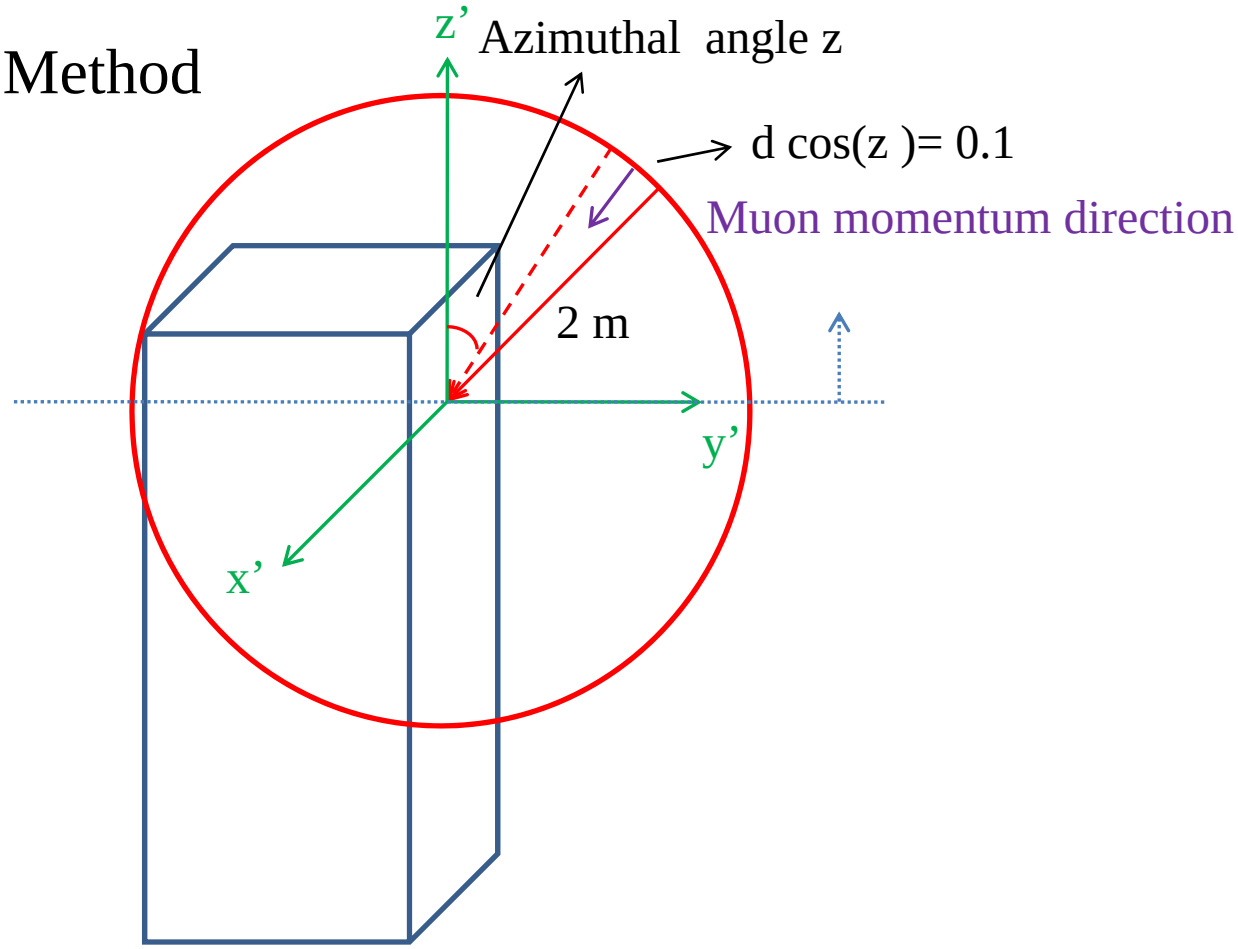
Simulation in GEANT4

- I study the GEANT4 by doing the simulation about the atmospheric muons which are secondary particles from cosmic ray interaction in atmosphere passing through the human body
- Calculating the energy absorption and discussing the effects in human body and analyzing the energy distribution for any kinds of particles
- We built a toy model of human and use GEANT4 to simulate the energy loss in human body by atmospheric muons

Simulation in GEANT4 (continued)

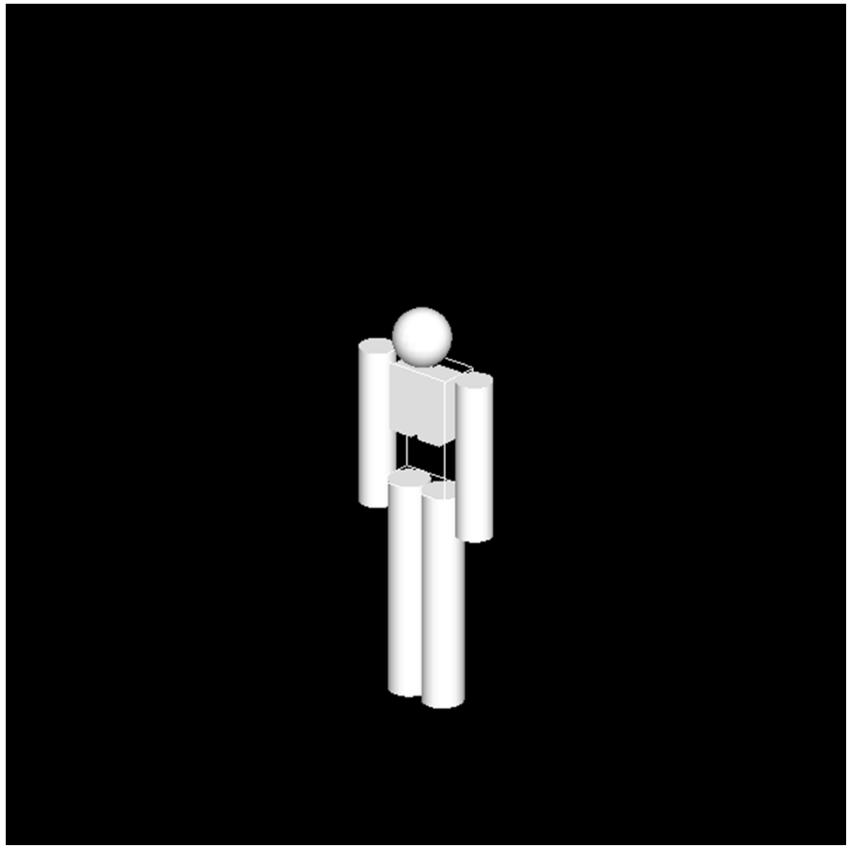
- Muon Flux
  - We applied the experimental data to simulate the generated cosmic muon

- Simulation Method



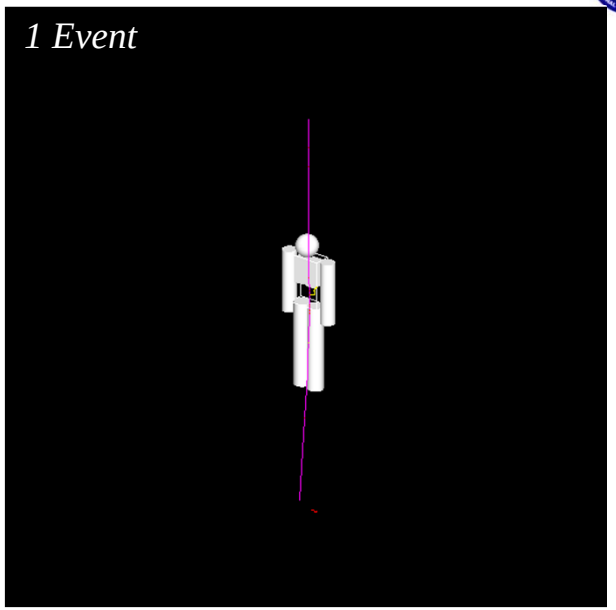
Simulation in GEANT4 (continued)

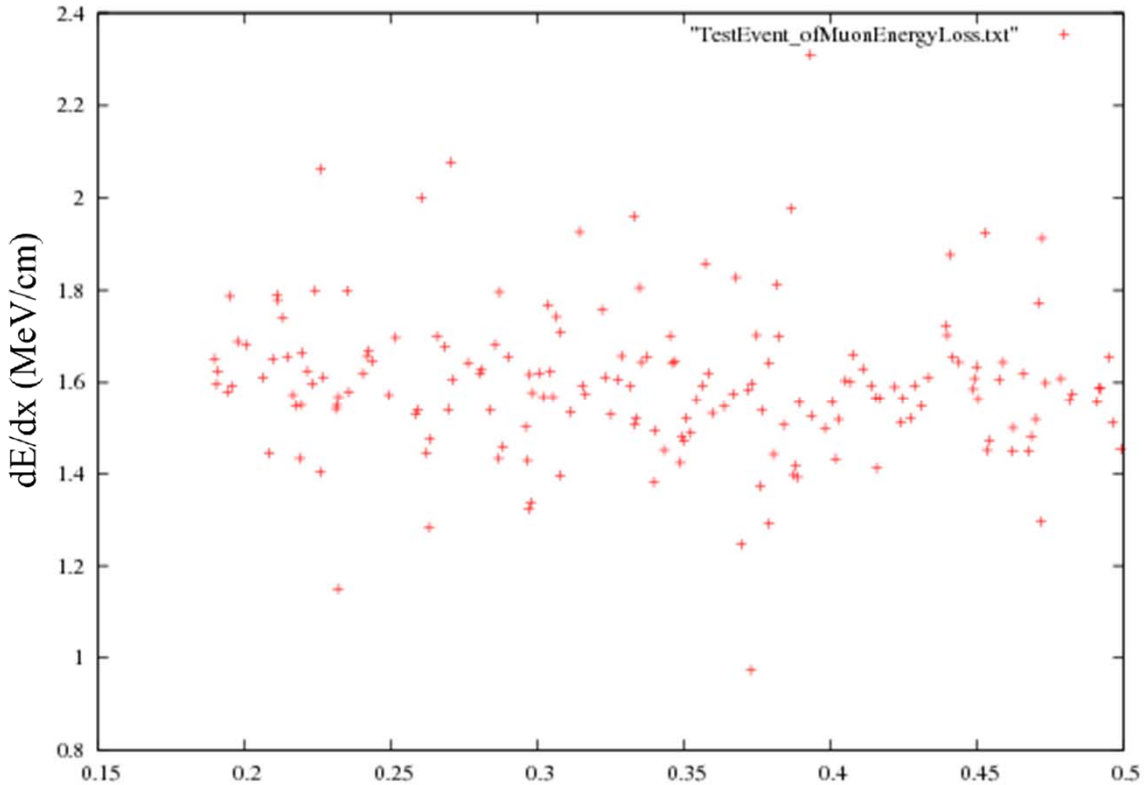
- The human body components of the toy model
  - Head (soft tissue)
  - Body (soft tissue)
  - Right and left hands (soft tissue)
  - Right and left legs (soft tissue)
  - Right and left lungs (lung material)
- Soft tissue(ρ=0.9869 g/cm^3):
  - H ~ 10%
  - C ~ 23%
  - O ~ 63%
- Lung material(ρ=0.2958 g/cm^3):
  - H ~ 10%
  - C ~ 10%
  - N ~ 2%
  - O ~ 75%

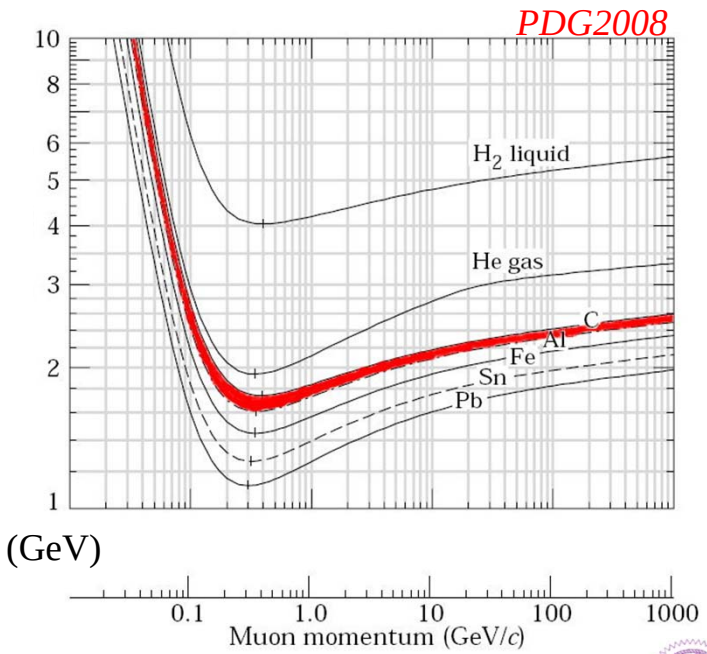


Some results of the present work

- Deposit of 1 muon event



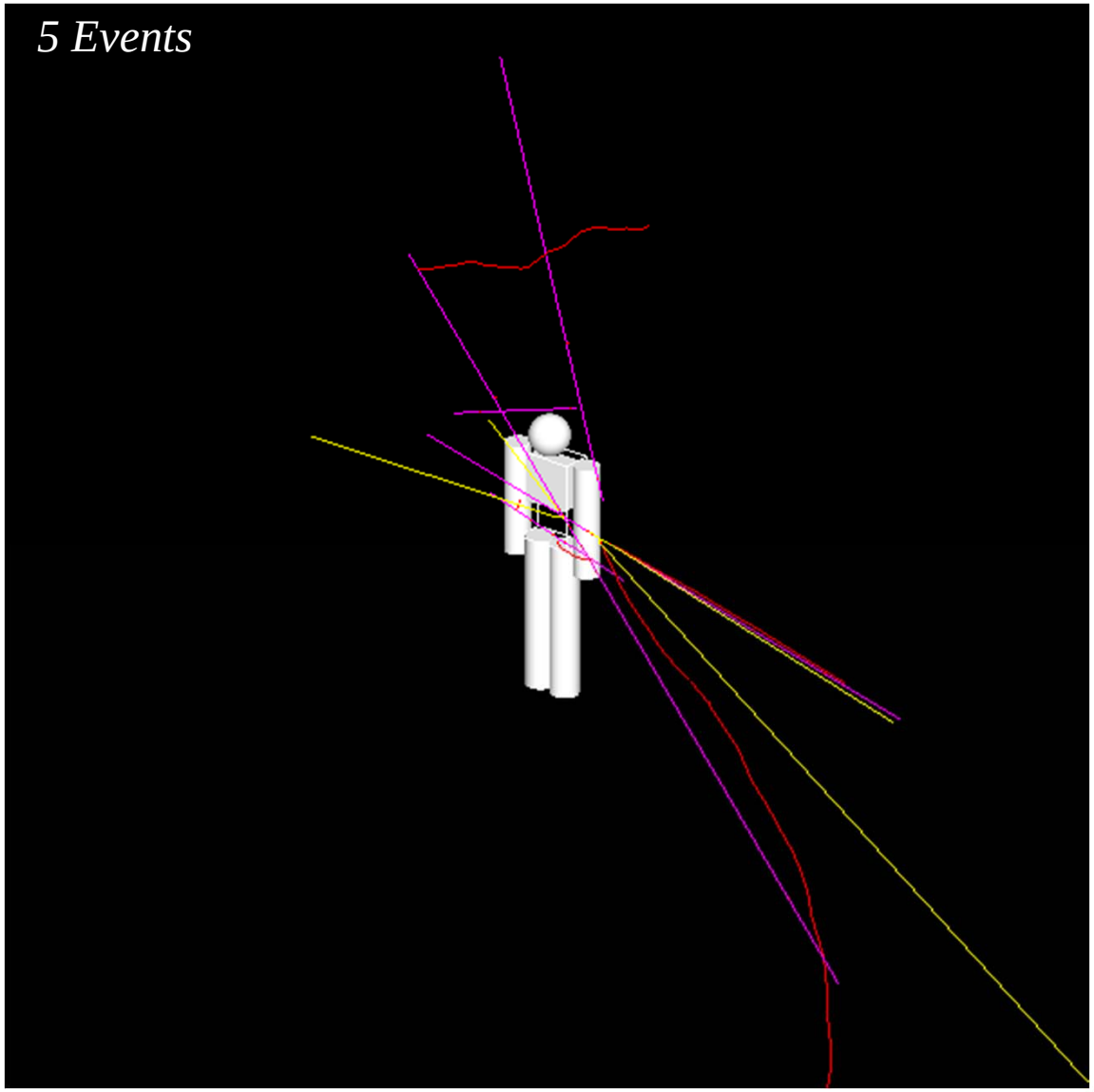




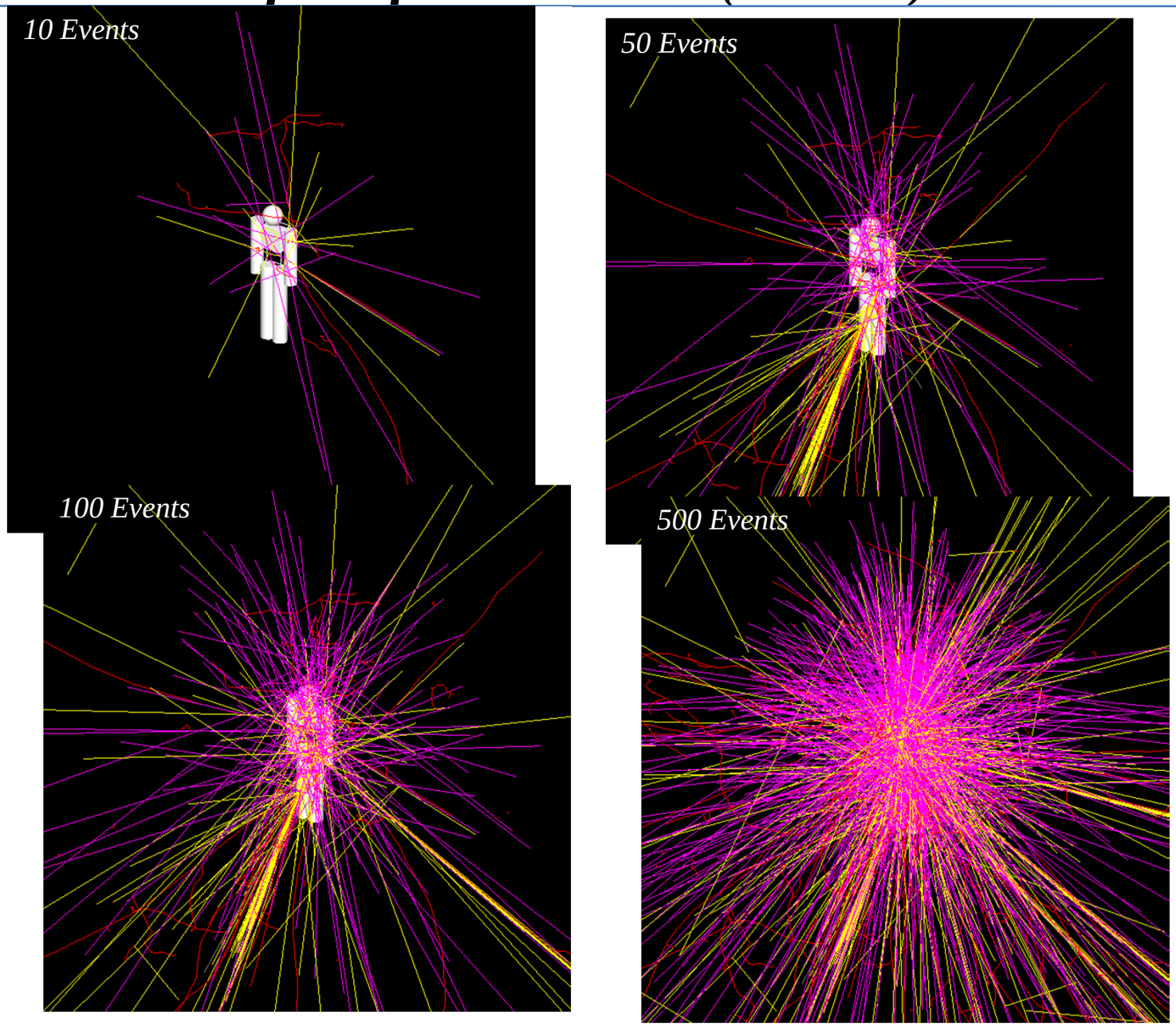
Some results of the present work (continued)

- Particle trajectory color definition

  - Muon: purple
  - Electron: red
  - Positron: blue
  - Photon: yellow
  - Others: gray

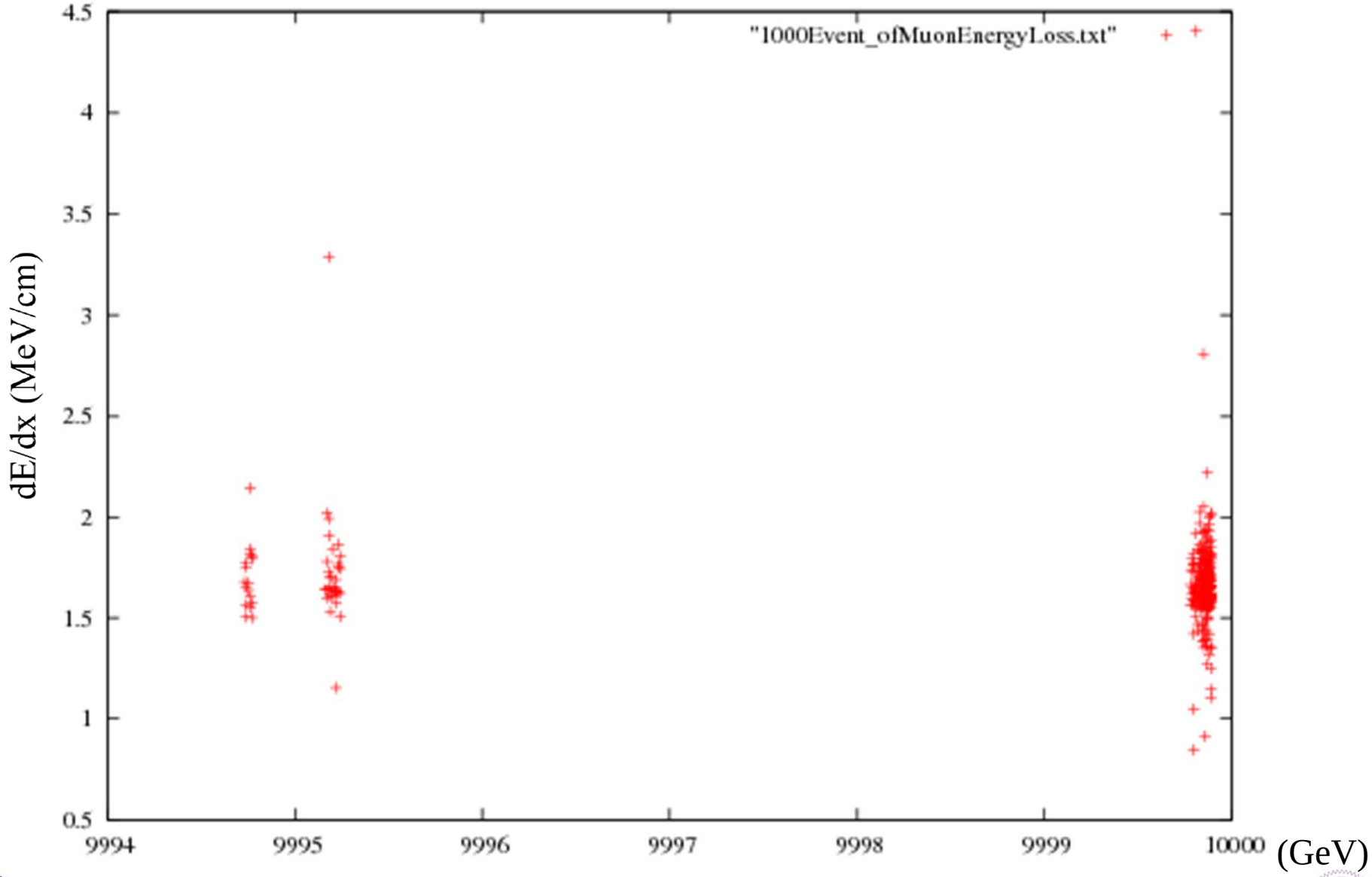


Some results of the present work (continued)



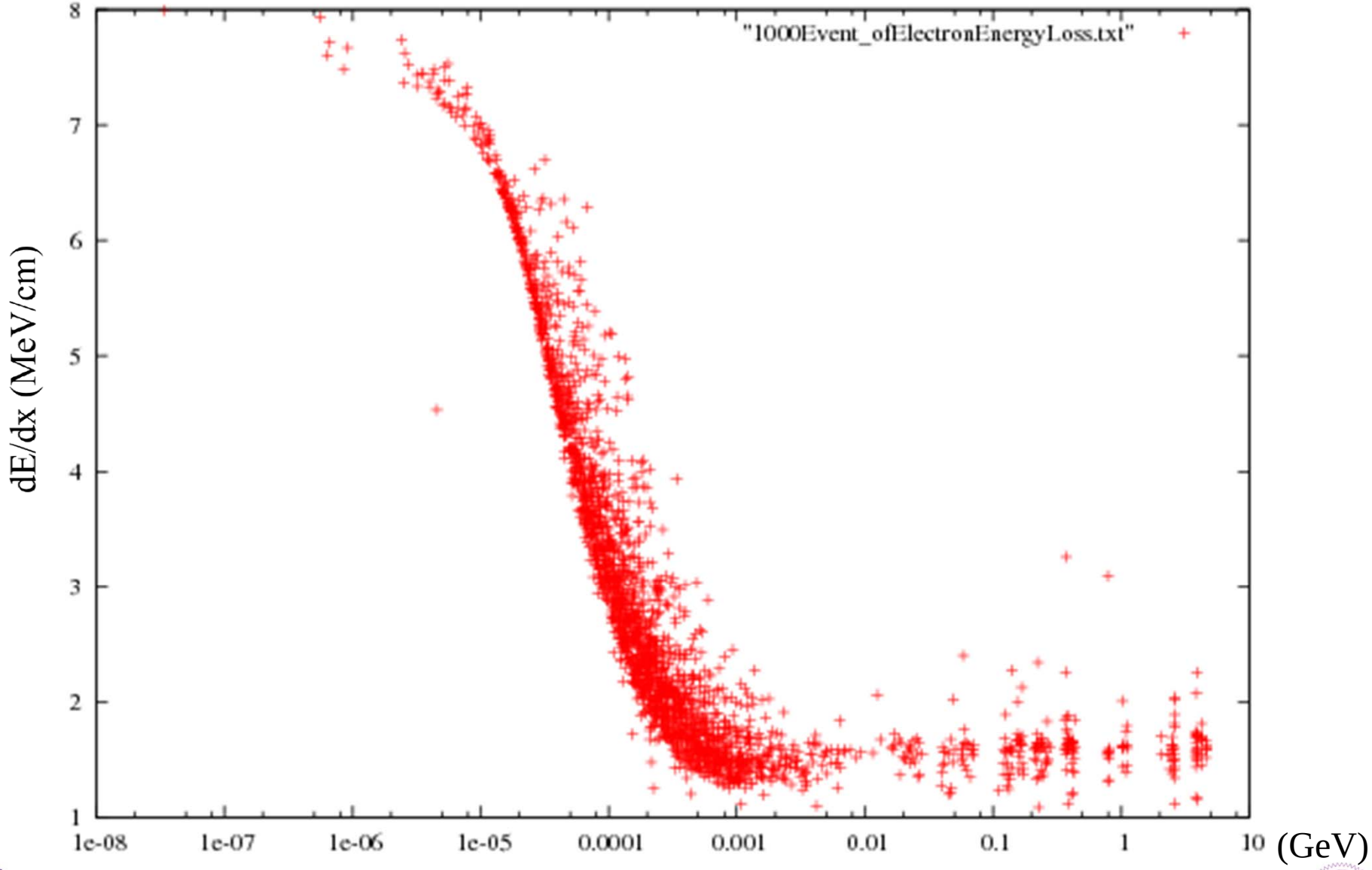
Some results of the present work (continued)

- Energy deposit of μ^-



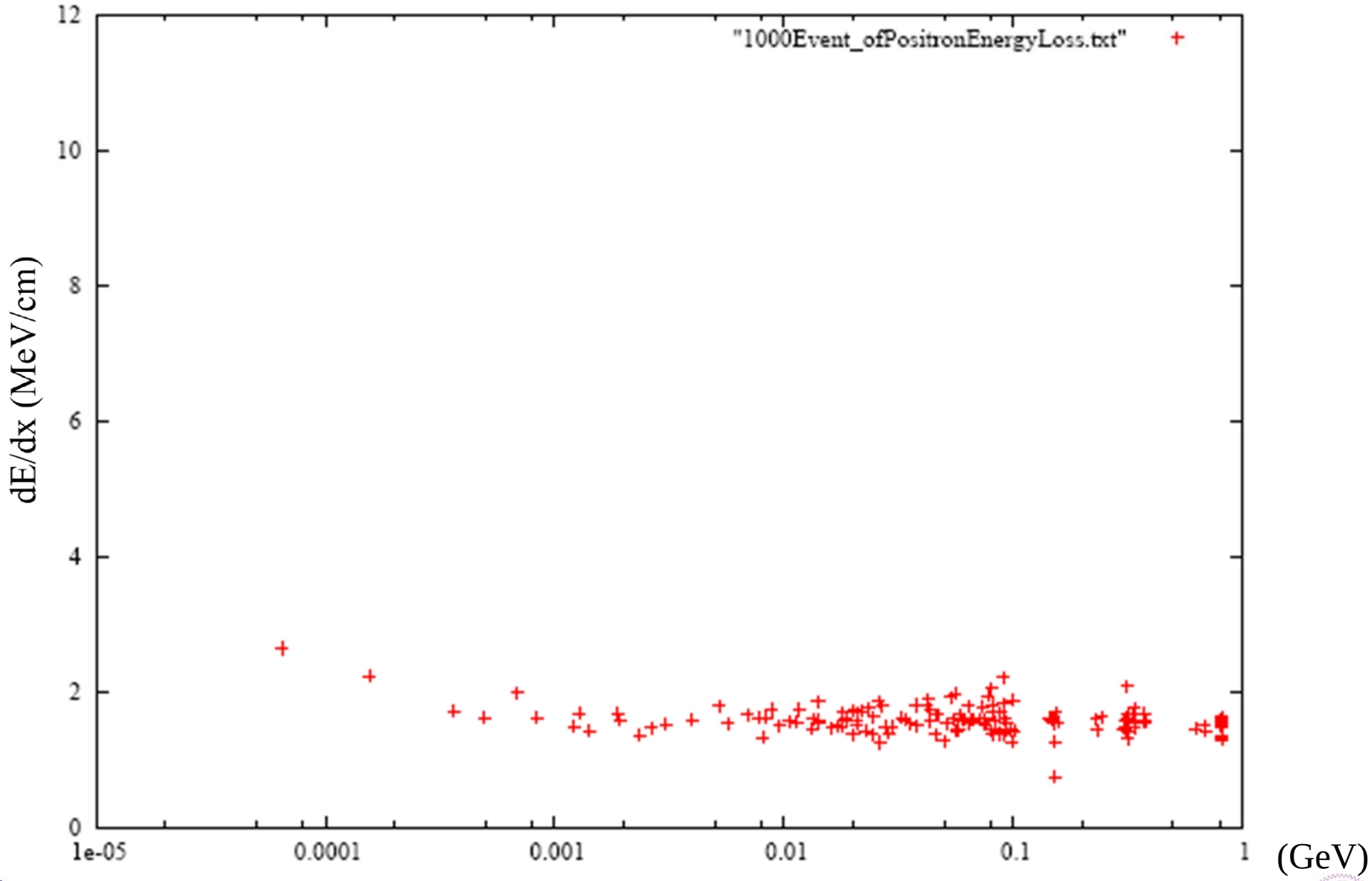
Some results of the present work (continued)

- Energy deposit of e^-



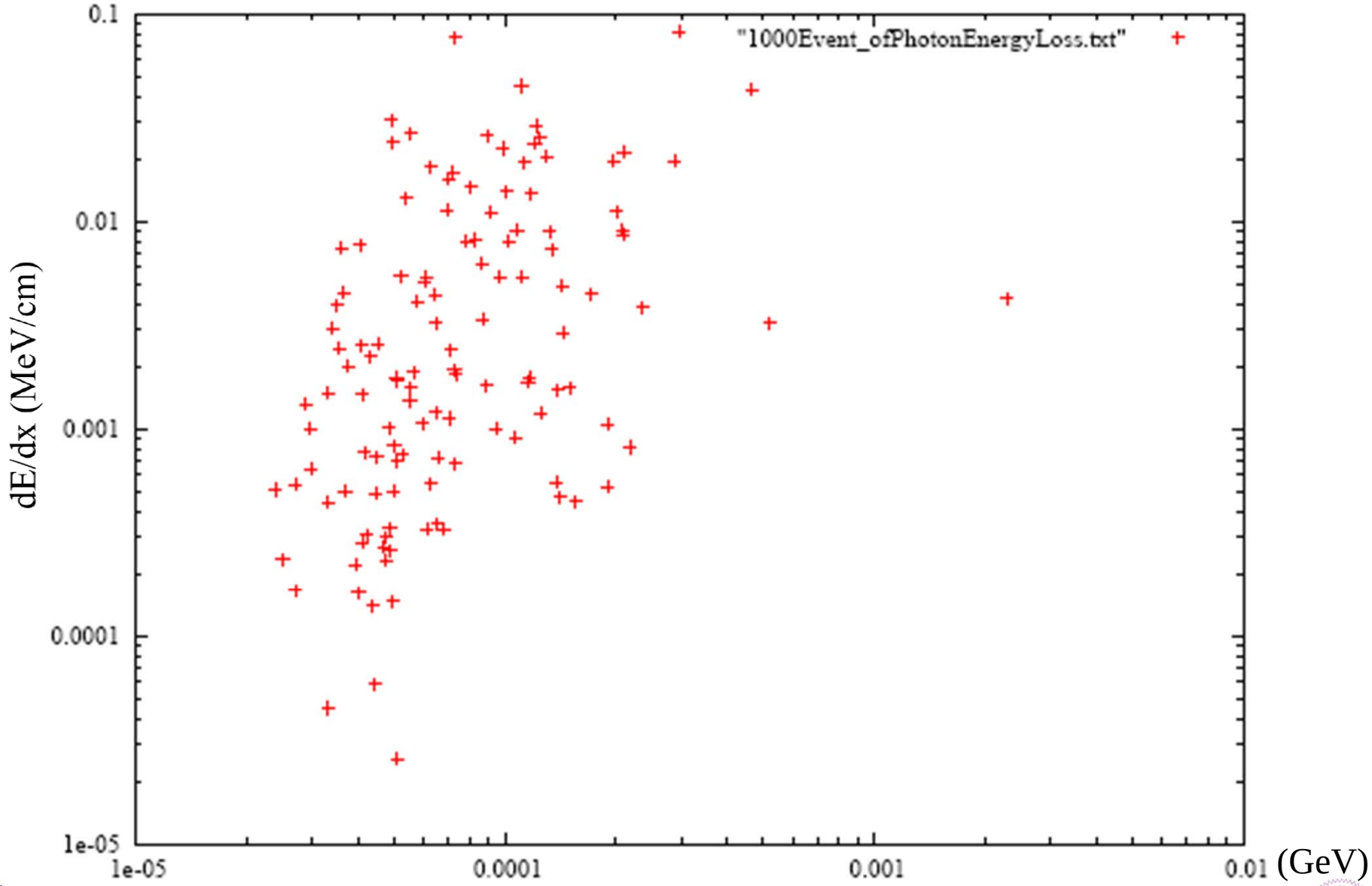
Some results of the present work (continued)

- Energy deposit of e^+



Some results of the present work (continued)

- Energy deposit of γ



Future work

- Calculate the energy deposit in the different volume component of human body
- Discuss what kind of the form of does the muon energy loss transfer to
- What is the energy distribution of different kind of particles produced by muon in the body
- Check if any hazards, like neutron, produced by the interaction of cosmic rays or not

Summary

- Muons are the most numerous charged particles at sea level
- The energy loss is small for high energy cosmic muons
- Most high energy muons will pass hourgh the human body
- The most particle produce is electron
- The photons, electrons and alittle positron would produce to take away the muon energy
- The energy of produced phton would be low
- The muon energy loss would have a tendency to be a stable nuber

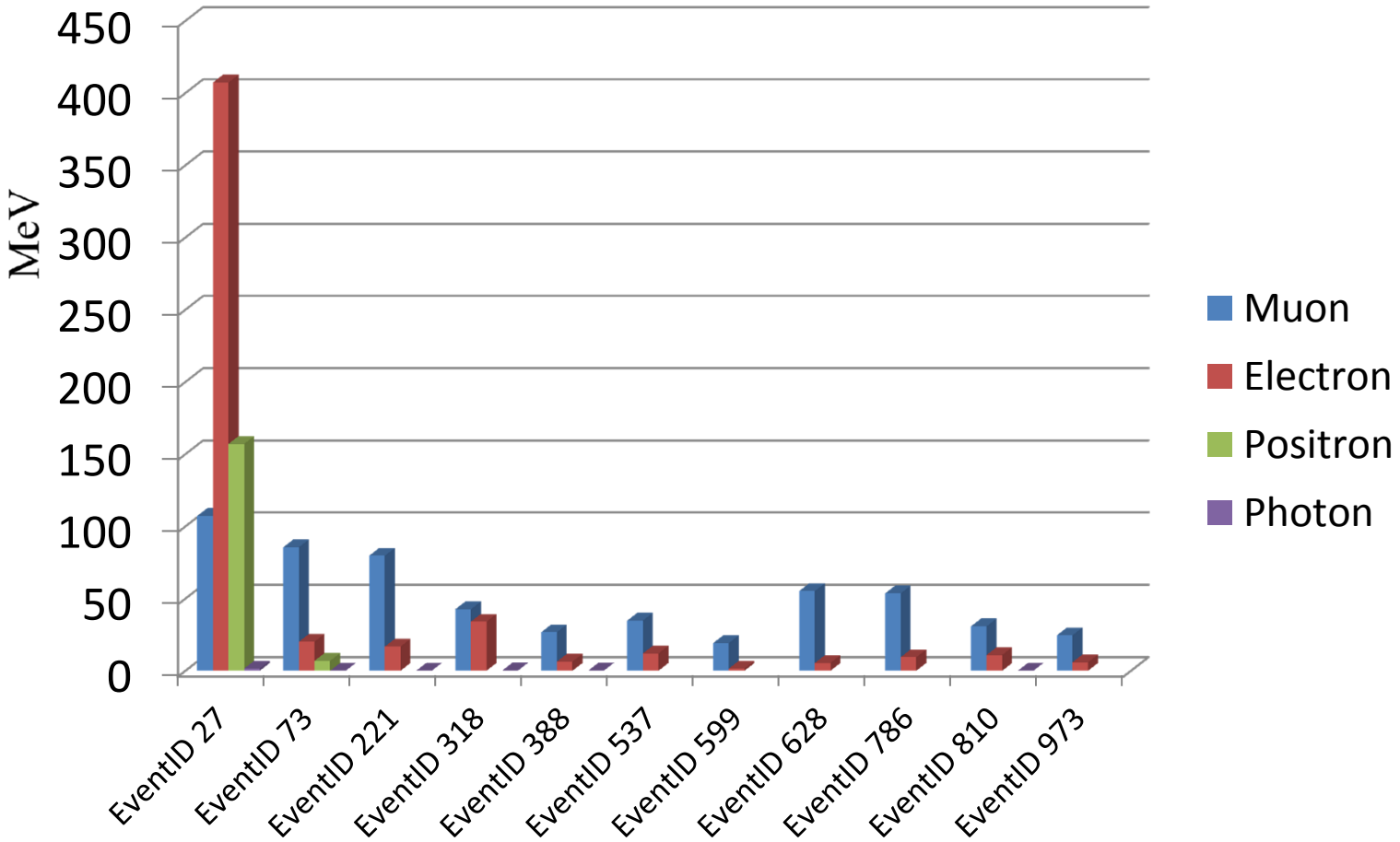
End

Thank you for listening!

New Progress

New progress

- First, I calculate the total energy deposit for γ, μ^-, e^- and e^+ in the human body



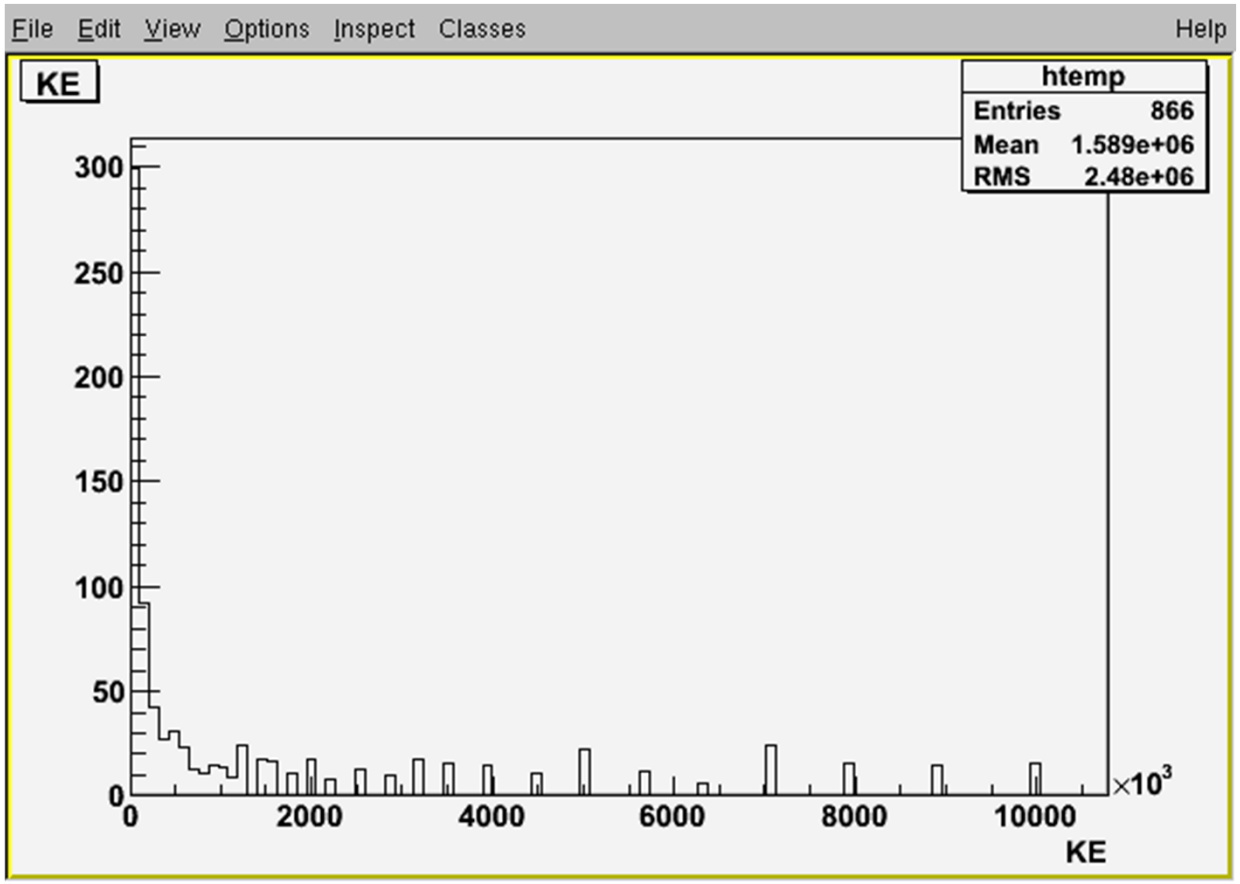
New progress (continued)

- Counting the particle # to plot the energy distribution

- But something is still wrong, I am checking what is the prob.
- A known prob.
  - When a particle passed through the boundary of the human body that the particle pass into the human body from one component to another, it might have double counting prob. in my program, so the # of μ^- energy distribution would be wrong

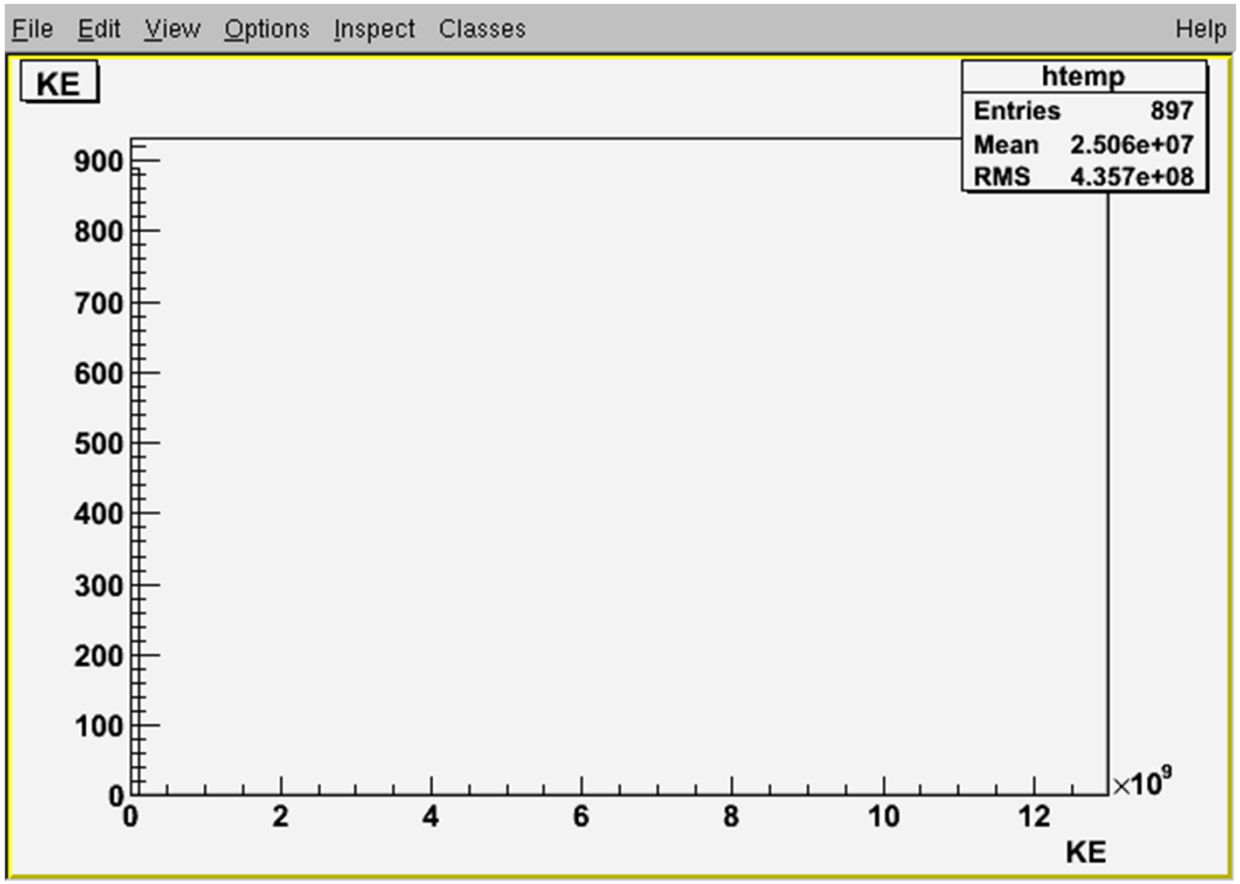
New progress (continued)

- Muon energy distribution



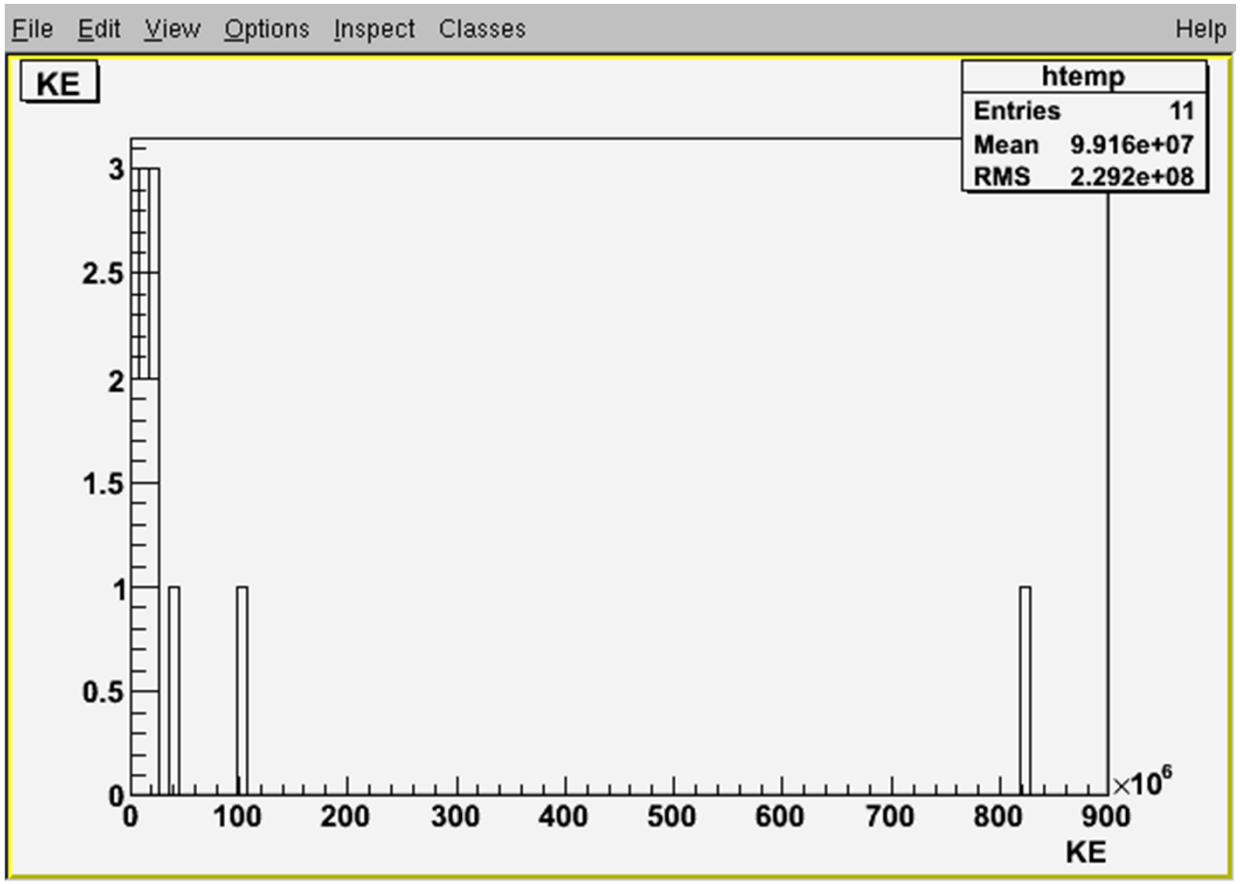
New progress (continued)

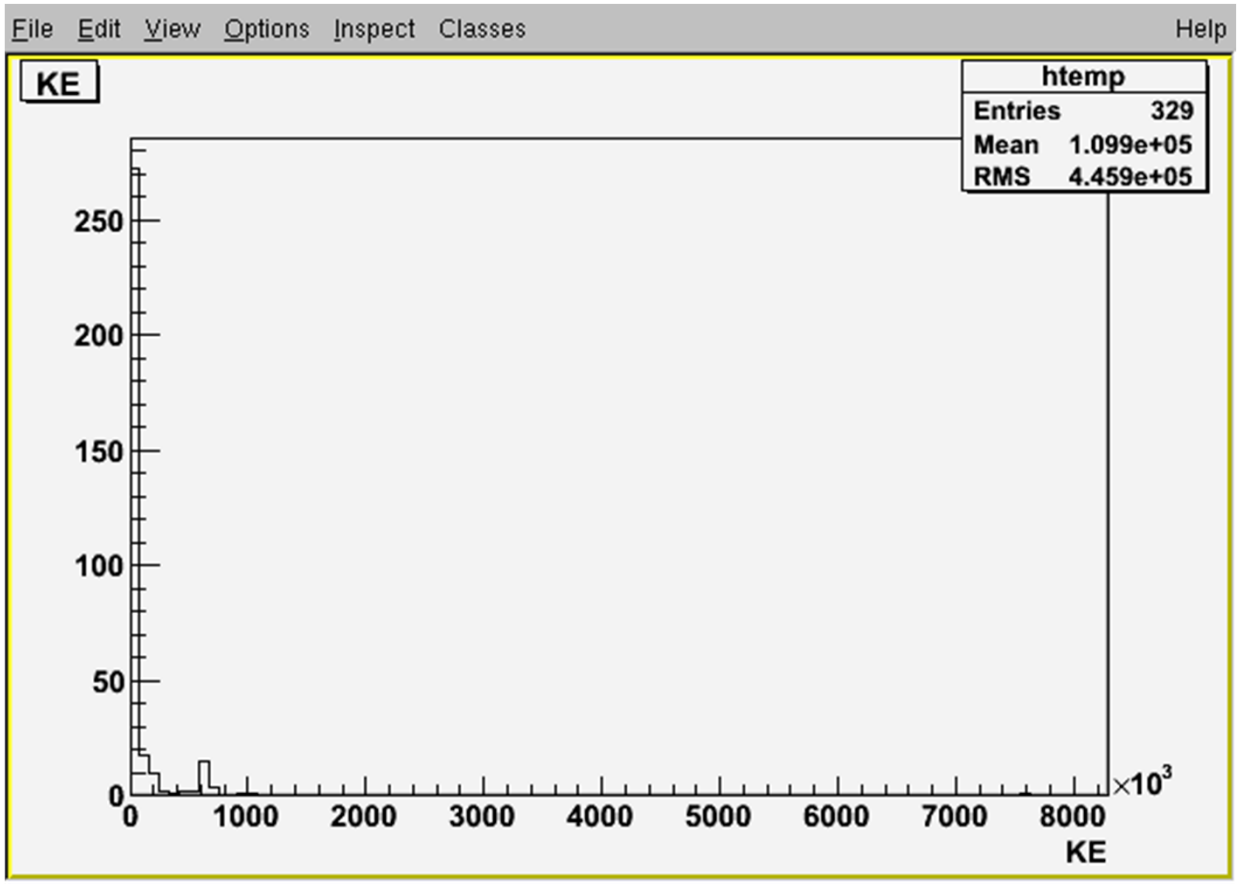
- Electron energy distribution



New progress (continued)

- Positron energy distribution



New progress (continued)

- Photon energy distribution